Model qualification report

Qwen2.5-0.5B-Instruct on Apple M5 Max (Metal). This sample uses FuseZap's own results on hardware we own. On your chip, the same report runs in your CI after every model, SDK or driver change.

MODEL	TARGET	REFERENCE
Qwen/Qwen2.5-0.5B-Instruct	Apple M5 Max · Metal	Hugging Face Transformers (PyTorch CPU, FP32)
FUSEZAP BUILD	**RUN DATE**	**TRIGGER**
commit a3d0cf2	16 October 2026	First qualification (no previous run)

QUALIFIED for FP16, FP32 and INT8 weights. FP8 qualified as weight-only. 4-bit only through AWQ/GPTQ checkpoints: naive INT4 failed the accuracy limit.

1. Correctness

CHECK	METHOD	RESULT	
Operator conformance	Every primitive and composed op vs NumPy and PyTorch, on four execution tiers (interpreter, NumPy, CPU, Metal)	5,739 passed, 144 skipped (axis-out-of-range cases)	PASS
Generation	64 greedy tokens from a fixed chat prompt vs the reference, FP32	Token-identical	PASS
Logits	Per-layer outputs vs the reference	Within atol 1e-3 (FP32)	PASS
Chat template	Rendered prompts vs Hugging Face for six tokenizers	Identical	PASS
Batch invariance	32 concurrent requests vs each request run alone, with chunked prefill	Bit-identical tokens	PASS
Preemption	Small KV block pool forcing preemption and recompute	All requests complete, none starve	PASS
API conformance	Official OpenAI Python client: chat, completions, streaming, 8 concurrent streams	All responses match engine output	PASS

2. Accuracy by number format

WEIGHTS	SCHEME	PERPLEXITY	VS FP16	AGREED LIMIT	
FP16	baseline	13.2014	—	—	PASS
INT8	per-channel, symmetric	13.1729	−0.22%	≤ +1%	PASS
FP8 E4M3	per-tensor scale, weight-only	13.2692	+0.51%	≤ +1%	PASS
INT4	group 128, asymmetric, round-to-nearest	15.7149	+19.04%	≤ +3%	FAIL
INT4	group 128, symmetric, round-to-nearest	17.6320	+33.56%	≤ +3%	FAIL

WikiText-2 raw test split, first 32,768 predictions, context 1024, stride 512, FP16 activations. Limits are agreed with each chip maker; these are examples. Action: ship 4-bit only from AWQ or GPTQ checkpoints, which FuseZap imports with dequantization matching the reference implementations bit for bit.

3. Performance

BENCHMARK	FP32	FP16	CHANGE VS LAST RUN
Decode, batch 1 (128-token prompt, 128 steps)	137.4 tok/s · p50 7.28 ms	149.2 tok/s · p50 6.67 ms	first run
Prefill, 128 tokens	2,954 tok/s	2,956 tok/s	first run
Prefill, 512 tokens	4,821 tok/s	5,323 tok/s	first run

CONTINUOUS BATCHING (FP32, 128 IN / 128 OUT)	BATCH 1	BATCH 8	BATCH 32
Decode tok/s	51.8	409.0	1,426.7
Aggregate tok/s, prefill included	51.4	384.1	1,140.8

On a quiet GPU with tuned kernels (decode beam width 4). A small model on Apple silicon: read this as proof the stack works, not as a speed claim for your chip. Later runs show the change against the previous report and flag any slowdown above the agreed threshold.

4. What this report does not cover yet

Larger models	7B and above are measured on NVIDIA H100 next; not part of this run.	PENDING
Comparison with another engine	Head-to-head against vLLM on the same hardware is in progress on H100.	PENDING
Long-run stability	A sustained-load run with latency drift and error rates is added from the H100 run onward.	PENDING

5. Reproduce

Conformance	`FZ_SLOW=1 uv run pytest -n 6`
Generation and logits	`uv run pytest tests/test_serve_model.py`
Batching	`uv run pytest tests/test_scheduler.py tests/test_kv_cache.py`
Accuracy	`uv run python bench/bench_ppl.py`
Performance	`uv run python bench/bench_decode.py` · `bench/bench_batch.py`
